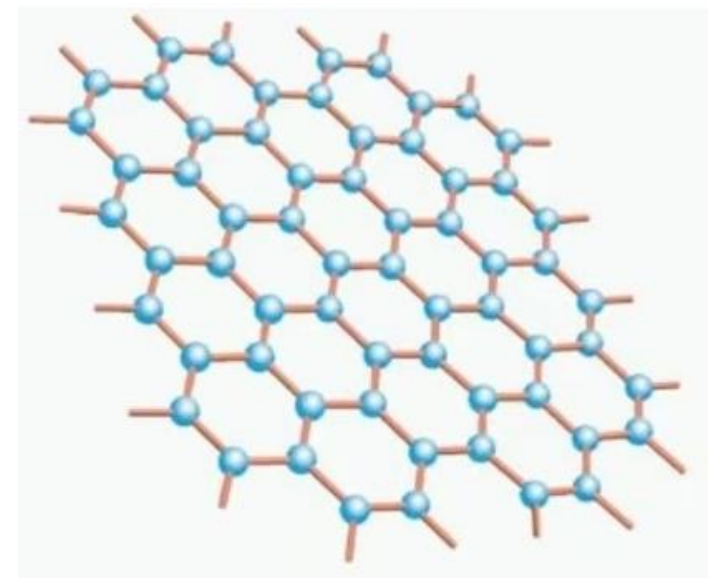
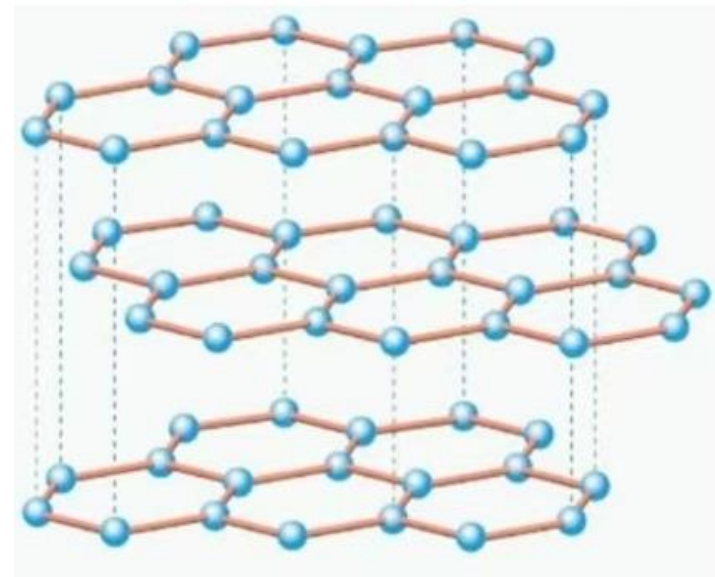




Nature loves hexagonal shape

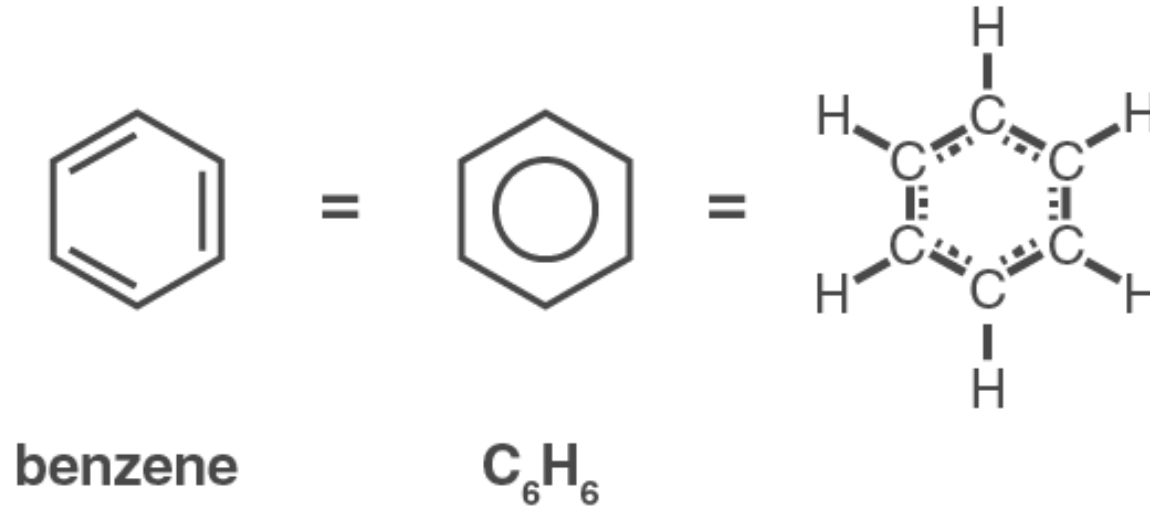


Nature loves hexagonal shape



Why does nature love hexagonal shape?

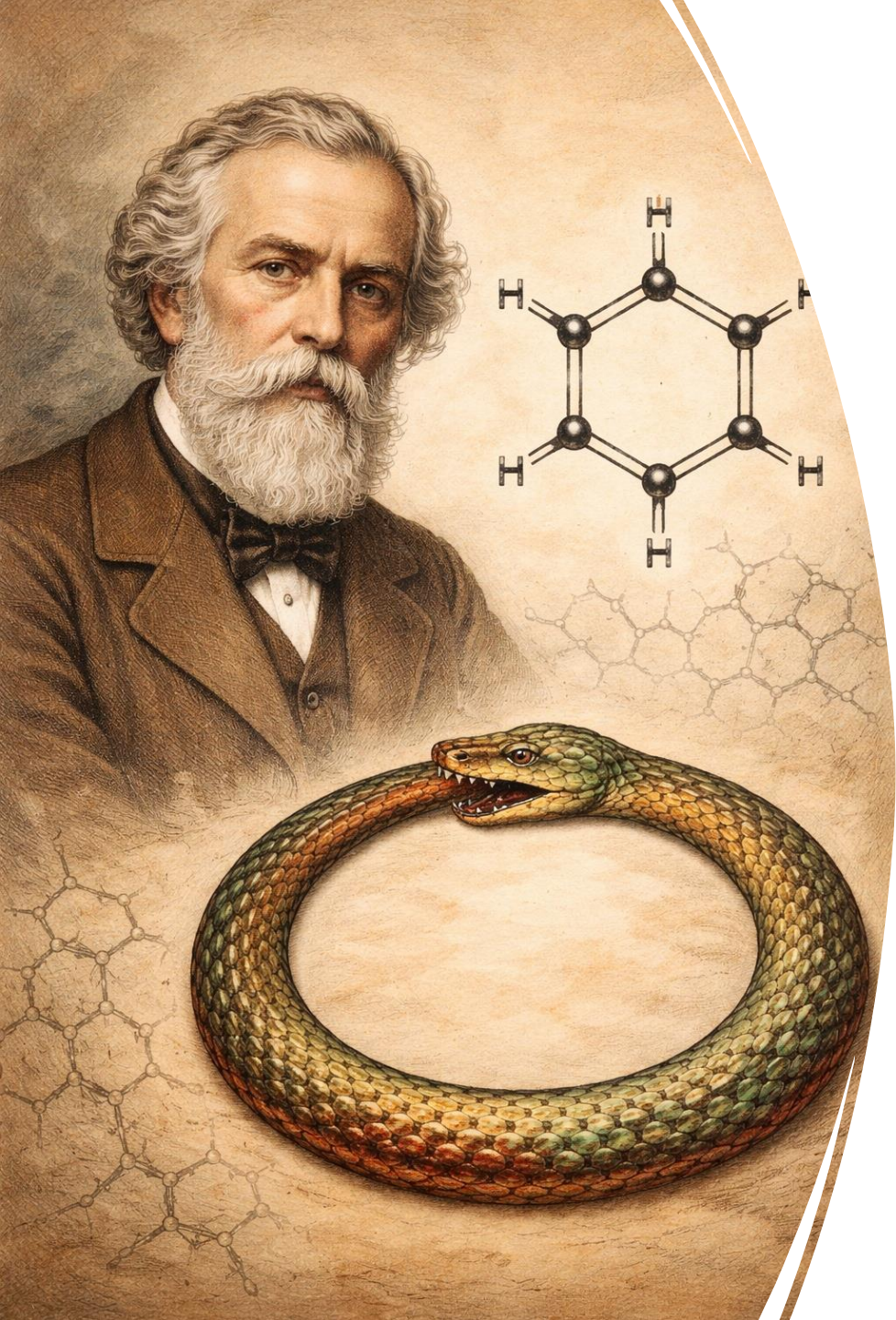
Seminal discovery



How the atoms in the mysterious compound benzene were arranged?

Organic Chemistry in the 19th century bloomed

Massive expansion of chemical industries towards the end of 19th century



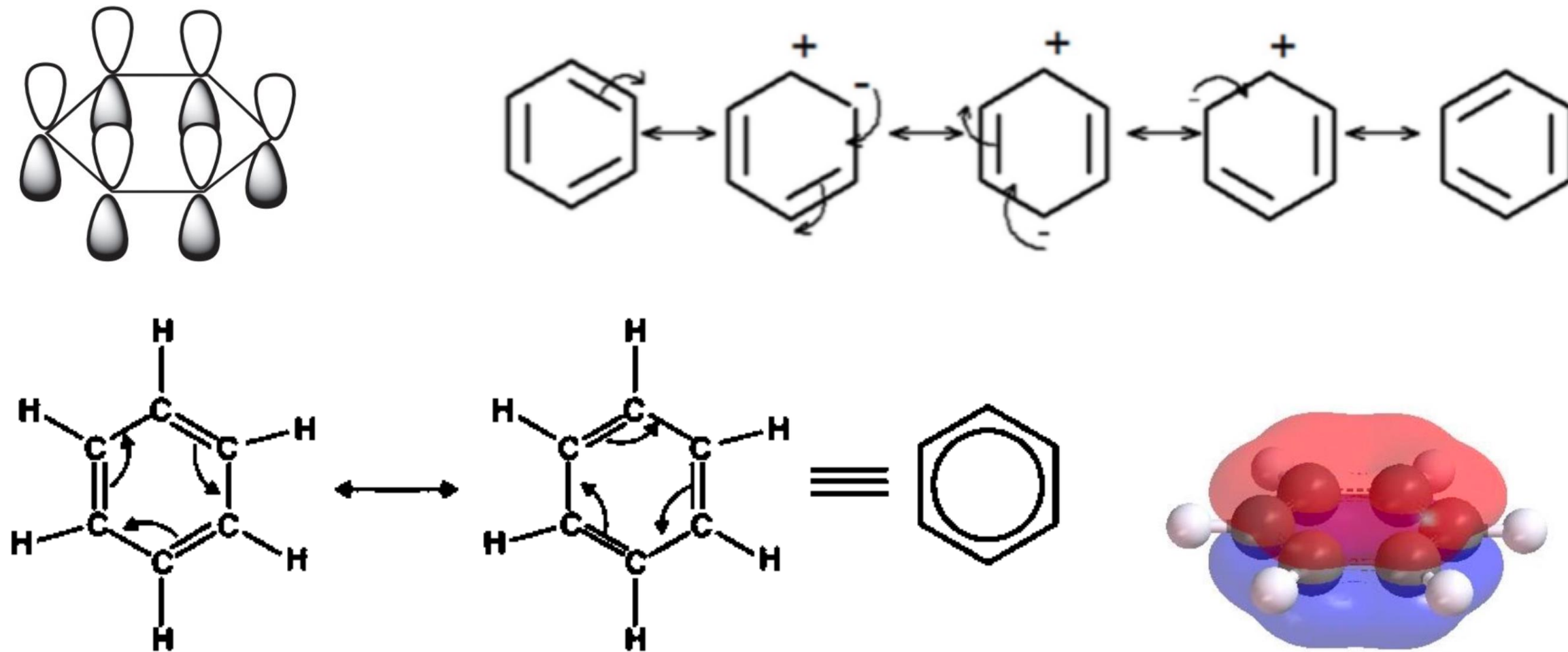
Kekulé, Benzene, and the Ouroboros

Resonance

Hyperconjugation

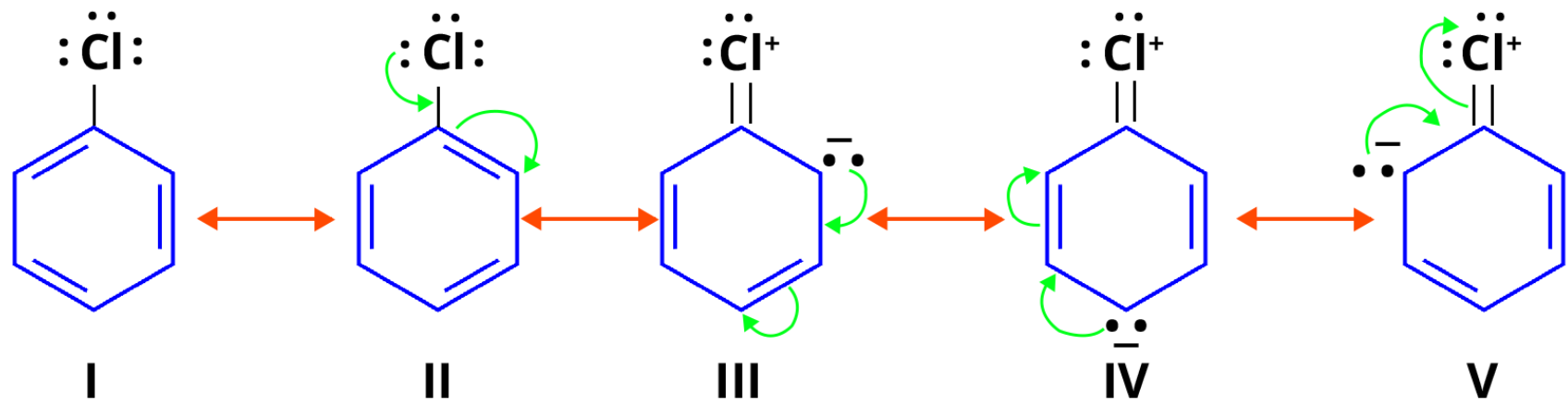
Inductive Effect

Resonance: Phenomena of the pi (π) bonds

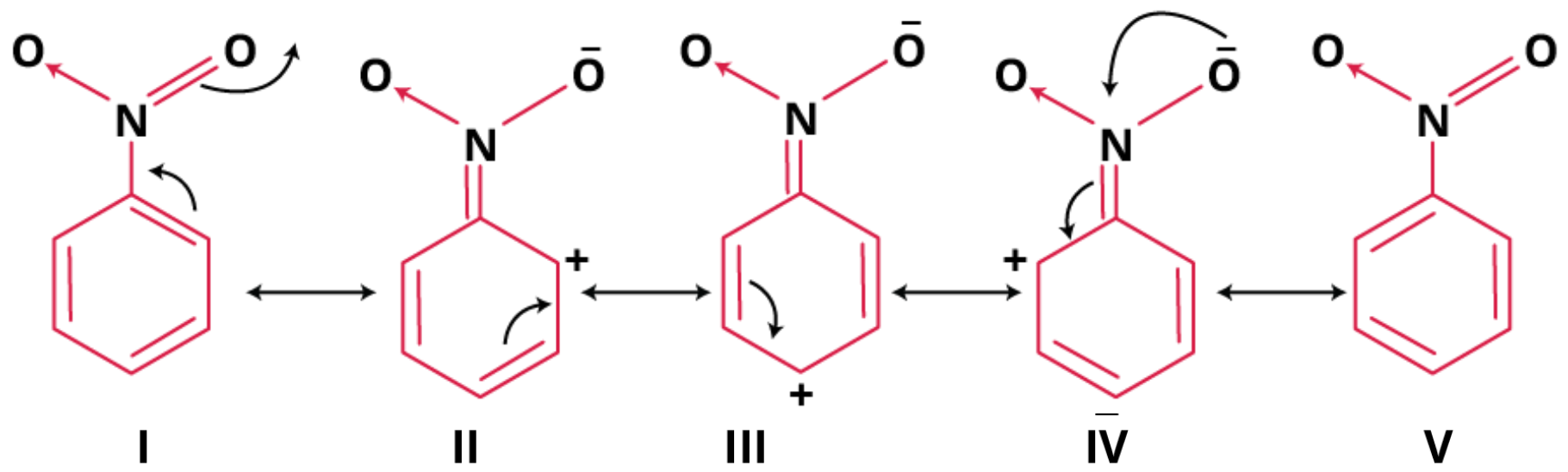


More the **number** of resonating structures, higher is the stability

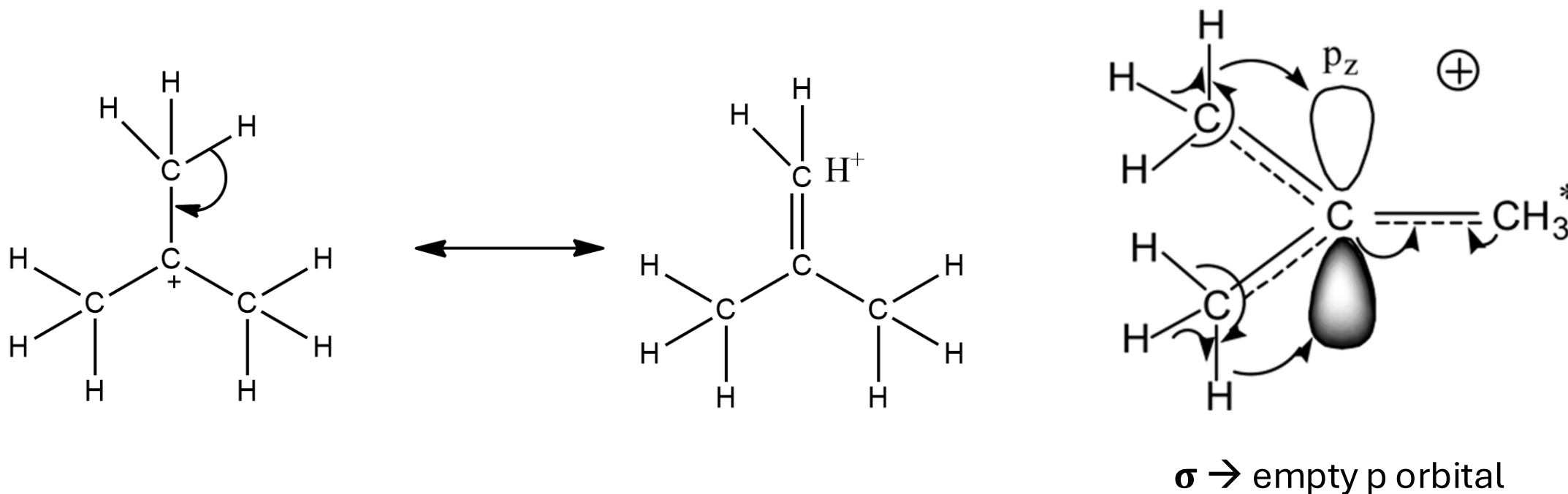
Electron Donating/ Releasing groups (EDG/ ERG)



Electron withdrawing groups (EWG)



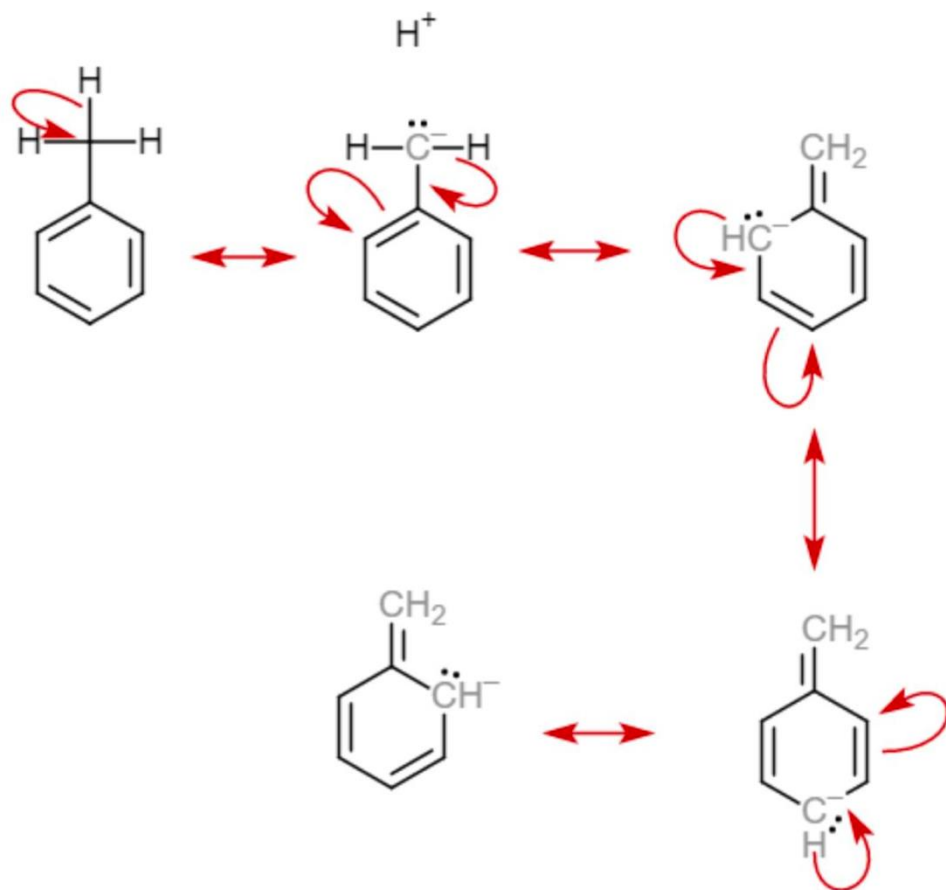
Hyperconjugation: Phenomena of the sigma bonds



More the **number** of hyperconjugation structures, higher is the stability

Resonance vs Hyperconjugation

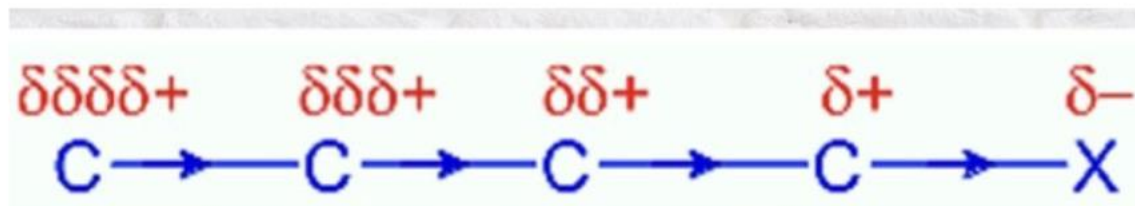
Toluene



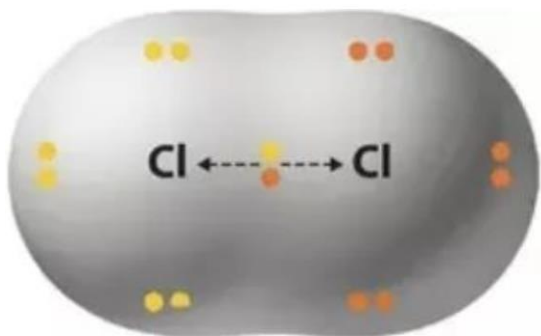
Sigma bonds are stronger than pi bonds

Hyperconjugation is more energy demanding than resonance

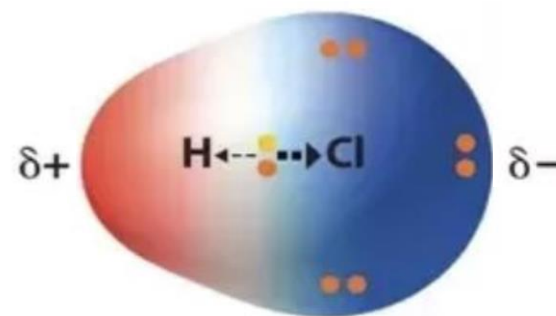
Inductive Effect



Inductive effect: Shift in electron density due to difference in electronegativity (leads to polarization)



CH₃-CH₃



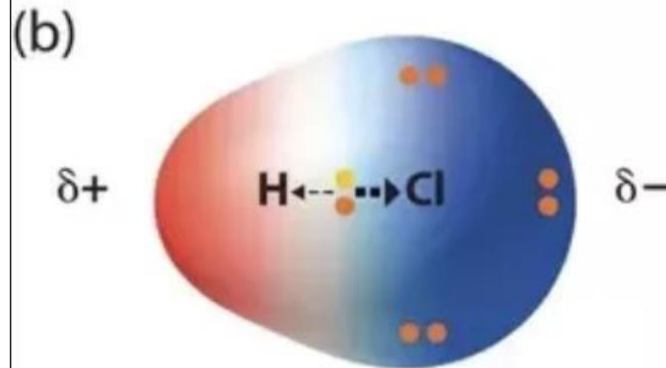
CH₃-Cl

Nature of chemical bonds and interactions



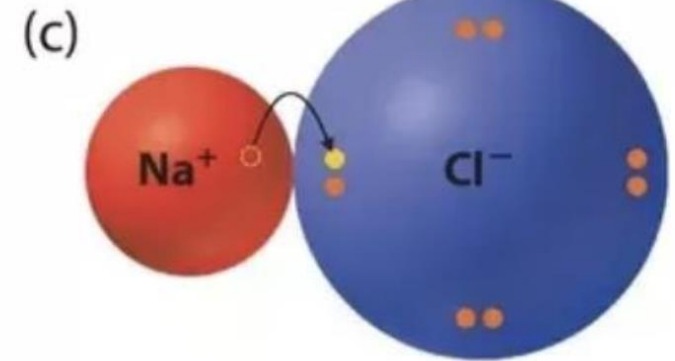
Nonpolar covalent bond

Bonding electrons shared equally between two atoms.
No charges on atoms.



Polar covalent bond

Bonding electrons shared unequally between two atoms.
Partial charges on atoms.



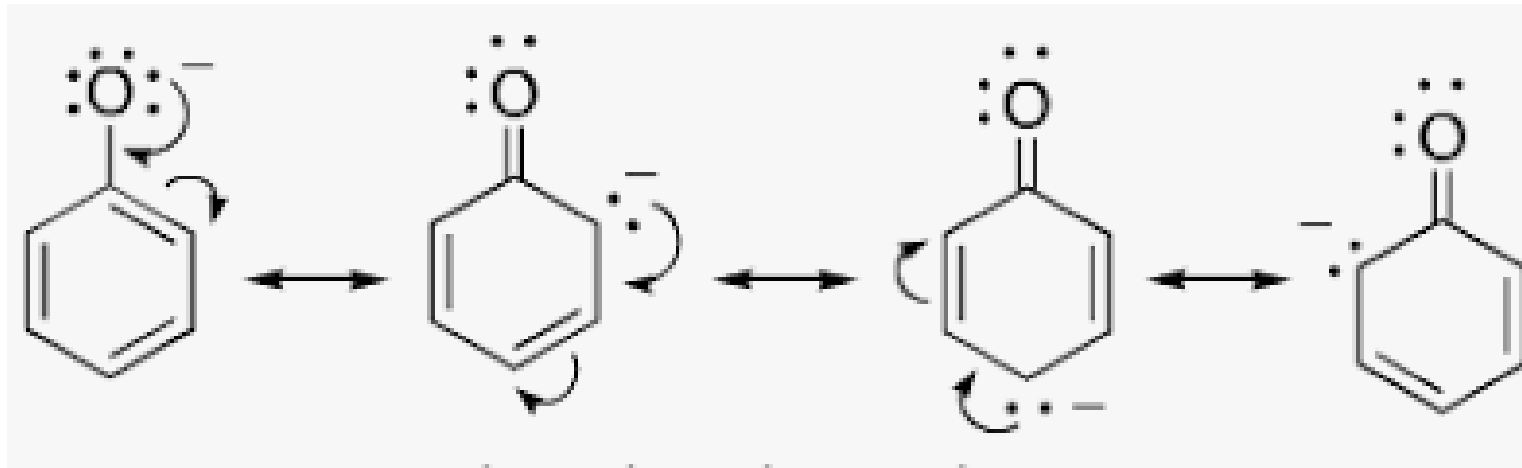
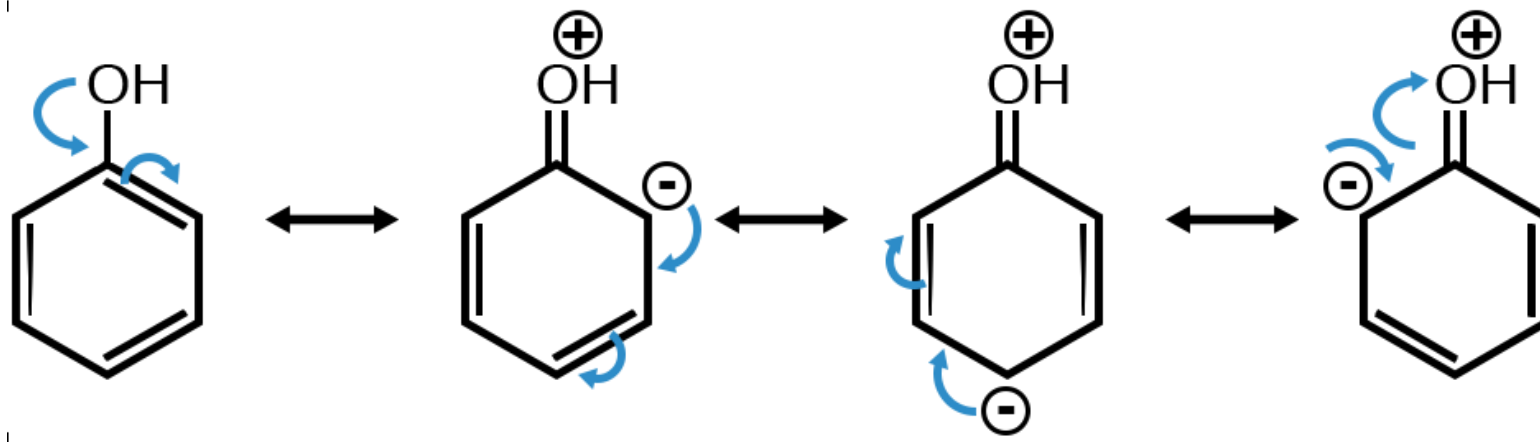
Ionic bond

Complete transfer of one or more valence electrons.
Full charges on resulting ions.

Tertiary butyl cation vs benzyl cation

Which one is more stable and why?

Electron Donating/Releasing groups (EDG/ERG)



Phenol vs Phenolate

Which one has stabilization more due to resonance effect

Polar Solvents

DMF

DMSO

Water

Acetone

Methanol

Isopropanol

Acetonitrile

Non-Polar Solvents

Hexanes/ PET Ether

Alkanes

Benzene

Toluene

Acetic acid

Chloroform

Diethyl ether

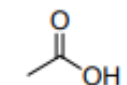
Ethyl acetate

Dichloromethane

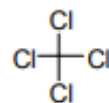
Polar Protic

Polar Aprotic

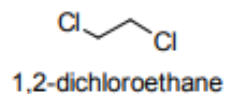
Identify the non-polar/ polar aprotic and polar protic solvents



acetic acid



carbon tetrachloride



1,2-dichloroethane



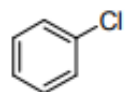
1,4-dioxane



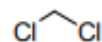
dimethyl sulfoxide



acetone



chlorobenzene



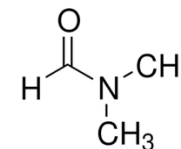
methylene chloride



tetrahydro furane



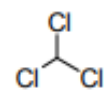
nitromethane



Dimethyl formamide



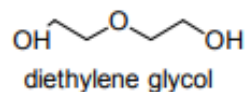
acetonitrile



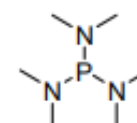
chloroform



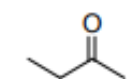
pyridine



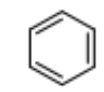
diethylene glycol



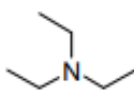
hexamethylphosphorous triamide (HMPT)



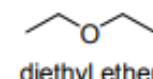
2-butanone



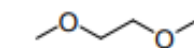
benzene



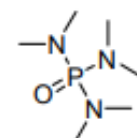
triethyl amine



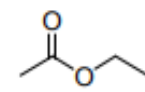
diethyl ether



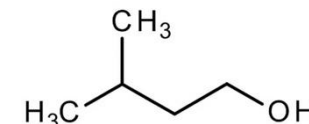
1,2-dimethoxyethane



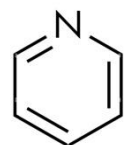
hexamethylphosphoramide (HMPA)



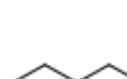
ethyl acetate



Isoamyl alcohol



pyridine



pentane



hexane



heptane



cyclohexane



methanol



ethanol



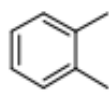
1-propanol



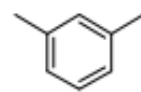
2-propanol



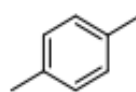
1-butanol



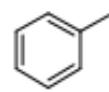
o-xylene



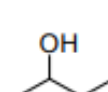
m-xylene



p-xylene



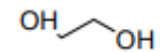
toluene



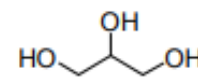
2-butanol



t-butyl alcohol



ethylene glycol



glycerine